les of Laplace transform pairs. $x(t) = 0$ for $t < 0^-$ and $T \geq 0$.

| | $x(t)$ | $\Longleftrightarrow$ | $\mathbf{X}(\mathbf{s}) = \mathcal{L}[x(t)]$ |
|---|---|---|---|
| 1 | $\delta(t)$ <- Impulse | $\Longleftrightarrow$ | $1$ |
| 1a | $\delta(t-T)$ Impulse train | $\Longleftrightarrow$ | $e^{-Ts}$ |
| 2 | $u(t)$ Unit step | $\Longleftrightarrow$ | $\frac{1}{s}$ |
| 2a | $u(t-T)$ Delayed Unit step | $\Longleftrightarrow$ | $\frac{e^{-Ts}}{s}$ |
| 3 | $e^{-at}u(t)$ Decay Expon. | $\Longleftrightarrow$ | $\frac{1}{s+a}$ |
| 3a | $e^{-a(t-T)}u(t-T)$ Expon shift | $\Longleftrightarrow$ | $\frac{e^{-Ts}}{s+a}$ |
| 4 | $t\,u(t)$ Ramp function | $\Longleftrightarrow$ | $\frac{1}{s^2}$ |
| 4a | $(t-T)u(t-T)$ | $\Longleftrightarrow$ | $\frac{e^{-Ts}}{s^2}$ |
| 5 | $t^2u(t)$ | $\Longleftrightarrow$ | $\frac{2}{s^3}$ |
| 6 | $te^{-at}u(t)$ | $\Longleftrightarrow$ | $\frac{1}{(s+a)^2}$ |
| 7 | $t^2e^{-at}u(t)$ | $\Longleftrightarrow$ | $\frac{2}{(s+a)^3}$ |
| 8 | $t^{n-1}e^{-at}u(t)$ | $\Longleftrightarrow$ | $\frac{(n-1)!}{(s+a)^n}$ |
| 9 | $\sin(\omega_0 t)u(t)$ sine functio | $\Longleftrightarrow$ | $\frac{\omega_0}{s^2+\omega_0^2}$ |
| 10 | $\sin(\omega_0 t+\theta)u(t)$ | $\Longleftrightarrow$ | $\frac{s\cdot\sin\theta+\omega_0\cos\theta}{s^2+\omega_0^2}$ |
| 11 | $\cos(\omega_0 t)u(t)$ cosine function | $\Longleftrightarrow$ | $\frac{s}{s^2+\omega_0^2}$ |
| 12 | $\cos(\omega_0 t+\theta)u(t)$ | $\Longleftrightarrow$ | $\frac{s\cdot\cos\theta-\omega_0\sin\theta}{s^2+\omega_0^2}$ |
| 13 | $e^{-at}\sin(\omega_0 t)u(t)$ | $\Longleftrightarrow$ | $\frac{\omega_0}{(s+a)^2+\omega_0^2}$ |
| 14 | $e^{-at}\cos(\omega_0 t)u(t)$ | $\Longleftrightarrow$ | $\frac{s+a}{(s+a)^2+\omega_0^2}$ |
| 15 | $2e^{-at}\cos(bt-\theta)u(t)$ | $\Longleftrightarrow$ | $\frac{e^{j\theta}}{s+a+jb}+\frac{e^{-j\theta}}{s+a-jb}$ |
| 15a | $e^{-at}\cos(bt-\theta)u(t)$ | $\Longleftrightarrow$ | $\frac{(s+a)\cos\theta+b\sin\theta}{(s+a)^2+b^2}$ |
| 16 | $\frac{2t^{n-1}}{(n-1)!}e^{-at}\cos(bt-\theta)u(t)$ | $\Longleftrightarrow$ | $\frac{e^{j\theta}}{(s+a+jb)^n}+\frac{e^{-j\theta}}{(s+a-jb)^n}$ |

$x(t) = \sin(\omega_0 t)u(t)$

## Method 1: Full Derivation

Step 1: Laplace

$$X(s) = \int_0^\infty e^{-st}\sin(\omega_0 t)\,dt$$

Step 2: Euler's

$$\sin(\omega_0 t) = \frac{e^{j\omega_0 t}-e^{-j\omega_0 t}}{2j}$$

$$X(s) = \int_0^\infty e^{-st}\cdot\frac{e^{j\omega_0 t}-e^{-j\omega_0 t}}{2j}$$

Step 3: Split into 2 integral.

$$X(s) = \frac{1}{2j}\left[\int_0^\infty e^{-(s-j\omega_0)t}\,dt - \int_0^\infty e^{-(s+j\omega_0)t}\,dt\right]$$

Step 4: Exponential Integral

$$\int_0^\infty e^{-at}\,dt \to \frac{1}{a} \quad (a>0)$$

$$\int_0^\infty e^{-(s-j\omega_0)t}\,dt = \frac{1}{s-j\omega_0}$$

$$\int_0^\infty e^{-(s+j\omega_0)t}\,dt = \frac{1}{s+j\omega_0}$$

$$= X(s) = \frac{1}{2j}\left[\frac{1}{s-j\omega_0}-\frac{1}{s+j\omega_0}\right]$$

Step 5: Complete the equation.

$$\frac{1}{s-j\omega_0}-\frac{1}{s+j\omega_0} = \frac{(s+j\omega_0)-(s-j\omega_0)}{(s-j\omega_0)(s+j\omega_0)} = \frac{2j\omega_0}{s^2+\omega_0^2}$$

$$\frac{1}{\cancel{2j}}\cdot\frac{\cancel{2j}\,\omega_0}{s^2+\omega_0^2} \to \frac{\omega_0}{s^2+\omega_0^2}$$

## Method 2: Quick Property

Step 1: Cos transform

$$\mathcal{L}[\cos(\omega_0 t)u(t)] = \frac{s}{s^2+\omega_0^2}$$

Step 2: Relate cos <—> sine

$$\frac{d}{dt}\cos(\omega_0 t) = -\omega_0\sin(\omega_0 t)$$

$$\sin(\omega_0 t) = \frac{1}{\omega_0}\left[\frac{d}{dt}(\cos(\omega_0 t)\right]$$

Step 3: Laplace property

$$\mathcal{L}\left[\frac{dx(t)}{dt}\right] = sX(s) - x(0^-)$$

$x(t) = \cos(\omega_0 t)u(t)$ and $x(0^-) = 1$

$$\mathcal{L}\left[\frac{d}{dt}\cos(\omega_0 t)u(t)\right] = s\cdot\frac{s}{s^2+\omega_0^2} - 1$$

Step 4: Simplify

$$s\cdot\frac{s}{s^2+\omega_0^2} - 1 \longrightarrow \frac{s^2}{s^2+\omega_0^2} - 1 = \frac{s^2-(s^2+\omega_0^2)}{s^2+\omega_0^2} = \frac{-\omega_0^2}{s^2+\omega_0^2}$$

Step 5: Relate back to sine.

Since $\frac{d}{dt}(\cos(\omega_0 t)u(t)) = -\omega_0\sin(\omega_0 t)u(t)$

then

$$\mathcal{L}[\underset{-\omega_0}{-\omega_0}\sin(\omega_0 t)u(t)] = \frac{-\omega_0^{\cancel{2}}}{s^2+\omega_0^2}\cdot\frac{-1}{\cancel{\omega_0}}$$

$$\mathcal{L}[\sin(\omega_0 t)u(t)] = \frac{\omega_0}{s^2+\omega_0^2}$$

---

## Class Practice

### We can clean up X(s) Partial Fractions

$$X(s) = \frac{s^2-4s+3}{s(s+1)(s+3)}$$

$$s\cdot X(s) = \frac{A_1}{s}\cdot s + \frac{A_2}{(s+1)}\cdot s + \frac{A_3}{(s+3)}\cdot s$$

$$sX(s) = A_1 + \frac{sA_2}{(s+1)} + \frac{sA_3}{(s+3)}$$

if we set $s=0$:

$$X(s) = \frac{s^2-4s+3}{s(s+1)(s+3)}$$

$A_1 = 1$

$$sX(s) = A_1 = \frac{s^2-4s+3}{(s+1)(s+3)}\bigg|_{s=0} = 1$$

$$(s+1)X(s) = A_2 = \frac{s^2-4s+3}{s(s+3)}\bigg|_{s=-1} = \frac{(-1)^2-4(-1)+3}{-1(-1+3)} = \frac{8}{-2} = -4$$

$$(s+3)X(s) = A_3 = \frac{s^2-4s+3}{s(s+1)}\bigg|_{s=3} = \frac{(3)^2-4(3)+3}{3(3+1)} = \frac{24}{6} = 4$$

$A_1 = 1$, $A_2 = -4$, $A_3 = 4$

---

plaace transform properties for causal functions, $x(t) = 0$ for $t < 0$.

| Property | $x(t)$ | $\Longleftrightarrow$ | $\mathbf{X}(\mathbf{s}) = \mathcal{L}[x(t)]$ |
|---|---|---|---|
| 1. Multiplication by constant | $K\,x(t)$ | $\Longleftrightarrow$ | $K\,\mathbf{X}(s)$ |
| 2. Linearity | $K_1x_1(t)+K_2x_2(t)$ | $\Longleftrightarrow$ | $K_1\mathbf{X}_1(s)+K_2\mathbf{X}_2(s)$ |
| 3. Time scaling | $x(at),\quad a>0$ | $\Longleftrightarrow$ | $\frac{1}{a}\mathbf{X}\left(\frac{s}{a}\right)$ |
| 4.Time shift | $x(t-T)u(t-T)$ | $\Longleftrightarrow$ | $e^{-Ts}\mathbf{X}(s)$ |
| 5. Frequency shift | $e^{-at}x(t)$ | $\Longleftrightarrow$ | $\mathbf{X}(s+a)$ |
| 6. Time 1st derivative | $x' = \frac{dx}{dt}$ | $\Longleftrightarrow$ | $s\mathbf{X}(s)-x(0^-)$ |
| 7. Time 2nd derivative | $x'' = \frac{d^2x}{dt^2}$ | $\Longleftrightarrow$ | $s^2\mathbf{X}(s)-sx(0^-)-x'(0^-)$ |
| 8. Time integral | $\int_{0^-}^t x(\tau)d\tau$ | $\Longleftrightarrow$ | $\frac{1}{s}\mathbf{X}(s)$ |
| 9. Frequency derivative | $t\,x(t)$ | $\Longleftrightarrow$ | $-\frac{d}{ds}\mathbf{X}(s) = -\mathbf{X}'(s)$ |
| 10. Frequency integral | $\frac{x(t)}{t}$ | $\Longleftrightarrow$ | $\int_s^\infty \mathbf{X}(q)dq$ |
| 11. Initial value | $x(0)$ | $=$ | $\lim_{s\to\infty} s\mathbf{X}(s)$ |
| 12. Final value | $\lim_{t\to\infty} x(t) = x(\infty)$ | $=$ | $\lim_{s\to 0} s\mathbf{X}(s)$ |
| 13. Convolution | $x_1(t)*x_2(t)$ | $\Longleftrightarrow$ | $\mathbf{X}_1(s)\mathbf{X}_2(s)$ |

$$\mathcal{L}[\cos(\omega_0 t)u(t)] = \frac{s}{(s-j\omega_0)(s+j\omega_0)}$$

$$X(s) = \frac{N(s)}{D(s)} \longrightarrow X(s) = \frac{s}{s^2+\omega_0^2}$$

$$= \frac{s}{(s+j\omega_0)(s-j\omega_0)}$$

---

$x(t) - \sin\omega_0 t)u(t)$

### 2nd Derivation

$$x'' = \frac{d^2x}{dt^2} \longleftrightarrow s^2X(s) - sx(0^-) - x'(0^-)$$

a) $x'(t) = \frac{d}{dt}[\sin(\omega_0 t)u(t)]$

$$= \omega_0\cos(\omega_0 t)u(t) + \sin(\omega_0 t)\,d[u(t)]$$

$$= \omega_0\cos(\omega_0 t)u(t) + \underline{\sin(\omega_0 t)\delta(t)}$$

$$= \omega_0\cos(\omega_0 t)u(t) + 0$$

$$\sin(\omega_0 t)\delta(t) = \sin(0)\delta(t) = 0$$

Second derv. of $x(t)$

$$x''(t) = \frac{d}{dt}[x'(t)] = \frac{d}{dt}[\omega_0\cos(\omega_0 t)u(t)]$$

$$= -\omega_0^2\sin(\omega_0 t)u(t) + \omega_0\overset{=1}{\cancel{\cos(\omega_0 t)}}\delta(t$$

$$= -\omega_0^2\sin(\omega_0 t)u(t) + \omega_0\delta(t)$$

$$\mathcal{L}[\sin(\omega_0 t)u(t)] = \frac{\omega_0}{s^2+\omega_0^2}$$

$\mathcal{L}[\delta(t)] = 1$ the Laplace of $x''(t)$

$$\mathcal{L}[x''(t)] = -\omega_0^2\left(\frac{\omega_0}{s^2+\omega_0^2}\right) + \omega_0$$

$$= s^2X(s) - sx(0^-) - x'(0^-)$$

---

## In special cases

Extreme cases:

$x(0^+)$: $t = 0^-$

$x(\infty)$: settled

corresponds

$t \to 0 \quad s \to \infty$

$t \to \infty \quad s \to 0$

---

### Extreme Case #1

$s \to 0 \quad t \to \infty$

$$\int_{0^-}^{0}\frac{dx(t)}{dt}e^{-st}\,dt = sX(s) - x(0^-)$$

$$\downarrow$$

$$\lim_{s\to 0}\left[\int_{0^-}^\infty \frac{dx(t)}{dt}\overset{=1}{\cancel{e^{-st}}}\,dt\right] = \lim[sX(s)] - x(0^-)$$

$$\downarrow$$

$$x(\infty) - \cancel{x(0^-)} = \lim_{s\to 0}[sX(s)] - \cancel{x(0^-)}$$

$$= \lim_{s\to 0}[sX(s)] = x(\infty)$$

this is a steady state

### Extreme Case 2.

$$\int_{0^-}^\infty \frac{dx(t)}{dt}e^{-st}\,dt = sX(s) - x(0^-)$$

$$= \lim_{s\to\infty}\left[\int_{0^-}^\infty \frac{dx(t)}{dt}e^{-st}\,dt\right] =$$

$$= \underbrace{\lim_{s\to\infty}\left[\int_{0^-}^{0^+}\frac{dx(t)}{dt}e^{-st}\,dt\right]}_{t\to 0\,=\,1} + \underbrace{\lim_{s\to\infty}\left[\int_{0^+}^{\infty}\frac{dx(t)}{dt}e^{-st}\,dt\right]}_{s\to\infty\,=\,0}$$

this come from the first

$$x(0^+) - \cancel{x(0^-)}$$

$$= \lim_{s\to\infty}[sX(s)] - \cancel{x(0^-)}$$

$$= \lim_{s\to\infty}[sX(s)] = x(0^+)$$

---

### Notes on Reconstruction: If the signal:

1. has a sampling rate below N.B.
2. has spectral copies that overlap (aliasing)
3. Doesn't allow a low pass filter to seprate the orginal.

Then this signal isn't Reconstructable.

If the signal is above the N.B, you might get reconstruction of the signal.

---

## Laplace Transform of the Impulse

Impulse

$$\mathcal{L}[\delta(t)] = \int_{0^-}^\infty \delta(t)e^{-st}\,dt = 1$$

Delayed Impulse

$$\mathcal{L}[\delta(t-T)] = \int_{0^-}^\infty \delta(t-T)e^{-st}\,dt = e^{-sT}$$

### Unit Step

$$\mathcal{L}[u(t)] = \int_{0^-}^\infty u(t)e^{-st}\,dt = -\frac{1}{s}e^{-st}\Big|_{0^-}^\infty = -\frac{1}{s}(0-1) = \frac{1}{s}$$

### Delayed Unit Step

$$\mathcal{L}[u(t-T)] = \int_{0^-}^\infty u(t-T)e^{-st}\,dt$$

$$= \int_T^\infty e^{-st}\,dt = -\frac{1}{s}e^{-st}\Big|_T^\infty = \frac{1}{s}e^{-Ts}$$

<- delayer term

Laplace Transform of the unit step

The delayed impulse can be revisited

$$\mathcal{L}[\delta(t-T)] = 1\cdot e^{-sT}$$

### Decaying Exponential

$$x(t) = e^{-at}u(t)$$

Multiplying by the unit step forces it to be causal

$$X(s) = \int_{0^-}^\infty e^{-at}u(t)e^{-st}\,dt = \int_{0^-}^\infty e^{-(s+a)t}\,dt$$

$$= \frac{e^{-(s+a)t}}{-(s+a)}\Big|_0^\infty = 0 - \frac{1}{-(s+a)} = \frac{1}{(s+a)}$$

---

### Time Scaling Property

$$\mathcal{L}[x(at)] = ?$$

Note that when $a<0$, the signal is no longer causal. So, we restrict ourselves to $a>0$.

$$\mathcal{L}[x(at)] = \int_{0^-}^\infty x(at)e^{-st}\,dt$$

Let $u = at \to du = a\,dt$

$$\mathcal{L}[x(at)] = \frac{1}{a}\int_0^\infty x(u)e^{-(s/a)u}\,du = \frac{1}{a}X\left(\frac{s}{a}\right) \quad a>0$$

---

### Signal Differentiation

$$\mathcal{L}\left[\frac{dx(t)}{dt}\right] = \int_{0^-}^\infty \underbrace{\frac{dx(t)}{dt}}_{g}\underbrace{e^{-st}}_{f}\,dt$$

This will use IBP:

$$uv - \int v\,du$$

$$= \overset{v}{x(t)}\overset{u}{e^{-st}}\Big|_{0^-}^\infty - \int_{0^-}^\infty \overset{v}{x(t)}\overset{du}{(-s)e^{-st}}$$

$$= -x(0^-) + s\int x(t)e^{-st}\,dt$$

$$\mathcal{L}\left[\frac{dx(t)}{d(t)}\right] = -x(0^-) + sX(s)$$

Say if we have a Repeated Diff.

$$\mathcal{L}\left[\frac{d^2x(t)}{dt^2}\right] = s[sX(s) - x(0^-)] - x'(0^-)$$

$$= s^2X_s - sx(0^-) - x'(0^-)$$

General Formula for $n^{th}$ derivation

$$\boxed{\mathcal{L}\left[\frac{d^nx(t)}{dt}\right] = s^nX(s) - s^{n-1}x(0^-) - s^{n-2}x'(0^-)\ldots - s^0x^{(n-1)}(0^-)}$$

★ Example: $\underbrace{\frac{d^2y(t)}{dt^2}} + \underbrace{\frac{dy(t)}{dt}} + \underline{y(t)} = x(t)$

$[s^2Y(s) - sy(0^-) - y'(0^-)] \to (s^2+s+1)Y(s)$

$[sY_s - y(0^-)] \to sy(0^-) - y'(0^-) \qquad = (s^2+s+1)Y(s) = X(s)$

$[Y(s)] \to -y(0^-)$

---

### Differentiation in the Freq. Domain.

$$X(s) = \int_{0^-}^\infty x(t)e^{-st}\,dt = \mathcal{L}[x(t)]$$

Let's deal with both sides

$$\frac{d}{ds}[X(s)] = \int_{0^-}^\infty \frac{d}{ds}[x(t)e^{-st}]\,dt$$

$$= \int_{0^-}^\infty -t\,x(t)e^{-st}\,dt$$

$$= \mathcal{L}[-t\,x(t)]$$

Example: $\mathcal{L}[t\,\underline{e^{-at}u(t)}]$ $\downarrow \frac{1}{s+a}$

So,

$$\mathcal{L}[te^{-at}u(t)] = -\frac{d}{ds}\left[\frac{1}{s+a}\right] = \frac{1}{(s+a)^2}$$

Freq. Domain Int.

$$\frac{x(t)}{t} \longleftrightarrow \int_s^\infty X(s')\,ds'$$

$$x(t) = \frac{\sin(t)}{t}u(t)$$

use this: $\mathcal{L}\left[\frac{\sin(t)}{t}u(t)\right] = \int_s^\infty \frac{1}{s'^2+1}\,ds'$

$$= \tan^{-1}(s')\Big|_s^\infty = \frac{\pi}{2} - \tan^{-1}(s)$$

---

### Inverse Laplace transform x(t)

$$X(s) = e^{-2s}\frac{s^2+s+3}{s^3+2s^2+s+2} = e^{-2s}Y(s)$$

Step 1: Factor the denominator

$$\underbrace{s^3+2s^2}+\underbrace{s+2} \to s^2(s+2)+1(s+2) = (s^2+1)(s+2)$$

$$\text{so } Y(s) = \frac{s^2+s+3}{(s+2)(s^2+1)}$$

Step 2: Partial Fraction

$$Y(s) = \frac{A}{(s+2)} + \frac{Bs+C}{(s^2+1)}\ (s+2)(s^2+1)$$

$(s+2)(s^2+1)$

$$s^2+s+3 = A(s^2+1) + (Bs+C)(s+2)$$

Let's find A.

Plug in $s=-2$ <— $(s+2)$

Step 3:

$$(-2)^2+(-2)+3 = A[(-2)^2+1]$$

$$= 4-2+3 = A(5) \to 5 = 5A = A = 1$$

Step 4: Find Bs+C

$$A(s^2+1) + (Bs+C)(s+2)$$

$$= As^2 + A + Bs^2 + 2Bs + Cs + 2C$$

$$= (A+B)s^2 + (2B+C)s + (A+2C)$$

$A=1$

$A+B=1$, $B=1-1=0$ | $2B+C=1$, $C=1-2(0)=1$ | $A+2C=3$, $1+2(1)=3$

$1+B=1$, $B=0$ | $2(0)+C=1$, $C=1$ | $1+2C=3$, $1+2(1)=3$

$$Y(s) = \frac{A}{s+2} + \frac{Bs+C}{(s^2+1)}$$

$$\frac{1}{s+2} + \frac{0+1}{(s^2+1)} = \frac{1}{s+2} + \frac{1}{(s^2+1)} = Y(s)$$

Step 5: Inverse Laplace Transform

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} = \frac{1}{s+2} = e^{-2t}u(t)$$

$$\sin(\omega_0 t)u(t) \longleftrightarrow \frac{\omega_0}{s^2+\omega_0^2} = \frac{1}{s^2+1} = \sin(t)u(t) \quad \omega_0 = 1$$

$$y(t) = e^{-2t}u(t) + \sin(t)u(t)$$

Step 6: Apply the time delay

$$X(s) = e^{-2s}Y(s)$$

time-shift property: $e^{-as}Y(s) \longleftrightarrow y(t-a)u(t-a)$

$a = 2$

$$x(t) = y(t-2)u(t-2) = e^{-2(t-2)}u(t-2) + \sin(t-2)u(t-2)$$

$$x(t) = \sin(t-2)u(t-2) + e^{-2(t-2)}u(t-2)$$

---

## Time Integration

$$\mathcal{L}\left[\int_0^t x(z)\,dz\right] = ?$$

Let $g(t) = \int_{0^-}^t x(z)\,dz$ and $g(0^-) = 0$

Now if $g(t) \longleftrightarrow G(s)$ then

$$X(s) = \mathcal{L}\left[\frac{dg(t)}{dt}\right] = sG(s) - g(0^-) = sG(s)$$

Therefore,

$$G(s) = \frac{X(s)}{s}$$

Remember this doesn't work when $s=0$.

If the lower limit is $-\infty$, we follow

$$\int_{-\infty}^t x(z)\,dz \longleftrightarrow \frac{X(s)}{s} + \frac{1}{s}\int_{-\infty}^{0^-}x(z)\,dz$$

---

## The Sinusoid

$$x(t) = \cos(\omega_0 t)u(t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega_0 t})u(t)$$

$$= X(s) = \frac{1}{2}\left[\mathcal{L}[e^{j\omega_0 t}u(t)] + \mathcal{L}[e^{-j\omega_0 t}u(t)]\right]$$

$$= \frac{1}{2}\left[\frac{1}{s-j\omega_0} + \frac{1}{s+j\omega_0}\right]$$

$$= \frac{1}{2}\left[\frac{(s+\cancel{j\omega_0})+(s-\cancel{j\omega_0})}{s^2+\omega_0^2}\right] = \frac{s}{s^2+\omega_0^2}$$

These denominators goes to zero under these conditions:

$s = j\omega_0 \quad s = -j\omega_0$

---

### ★ Inverse Laplace transform

$$X(s) = \frac{1}{(s+2)^3}$$

$\downarrow$

we can look at the table and see that this is similar to one of the transform

$$te^{-at}u(t) \longleftrightarrow \frac{1}{(s+a)^2}$$

$$= te^{-2t}u(t)$$

Also understand that $\frac{1}{s^3}$ follow the $t^2u(t)$ rule

$$\mathcal{L}[t^2u(t)] = \frac{2}{s^3},\ \frac{1}{2}\cdot\frac{2}{s^3} = \frac{1}{s^3} = \frac{1}{2}t^2u(t)$$

$$= \frac{1}{2}e^{-2t}t^2u(t)$$

---

## Partial Functions

$$X(s) = \frac{N(s)}{D(s)} = \frac{b_0 + b_1s + \ldots + b_ms^m}{a_0 + a_1s + \ldots + a_ns^n}$$

$m$ = # of zeros

$n$ = # of poles

$$C\left[\frac{(s-z_1)(s-z_2)\ldots(s-z_m)}{(s-p_1)(s-p_2)\ldots(s-p_n)}\right]$$

- $m<n$: Strictly proper <- most convenient
- $m=n$: Proper
- $m>n$: Improper

---

Note that as $t\to\infty$, this is underdamp making the poles unstable.

### Checking for underdamp.

Example:

$$X(s) = \frac{s+2}{s(s-1)}$$ <— If we apply the theorem

$$= \lim_{s\to 0}[sX(s)] = \lim_{s\to 0}\left[\frac{s+2}{s-1}\right] \to -2$$

This limit is misleading since the system is BIBO unstable.

$$x(t) = -2u(t) + \underline{3e^{t}}u(t)$$

pole on the right half plane.

Note that a system with more than one root at the origin are also unstable poles.

---

General note of Linearity

$T[\cdot]$ is linear: $T[ax_1 - bx_2] = aT[x_1] + bT[x_2]$

Shortcut: if you plug $x(t) = 0$ and get $y(t) \neq 0$ then its not linear.

Time Invariance: does the delaying input just delay the output, with everything else unchanged?

$y_2(t) = ax_1(t-t_0) \quad = \quad y(t-t_0) = bx_2(t-t_0)$

---

### Repeated Real Poles

$$X_1(s) = \sum_{i=1} \frac{A_i}{s-p_i}$$

Step 1: Given a strictly proper rational function $X(s)$

$$X(s) = X_1(s)X_2(s)$$

$$X_1(s) = \frac{N(s)}{(s-p_1)(s-p_2)\ldots(s-p_n)}$$

$$X_2(s) = \frac{1}{(s-p)^m}$$

this has an m-repeated pole at $s=p$

the repeated pole is not a pole of $X_1(s)$; $p \neq p_i$

Step 2: Partial fraction.

$$\frac{B_1}{(s-p)} + \frac{B_2}{(s-p)^2} + \ldots + \frac{B_m}{(s-p)^m}$$

$$X(s) = \frac{A_1}{(s-p_1)} + \frac{A_2}{(s-p_2)} + \ldots + \frac{A_n}{(s-p_n)} + \frac{B_1}{(s-p)} + \frac{B_2}{(s-p)^2} + \ldots + \frac{B_m}{(s-p)^m}$$

$$= \sum_{i=1}^n \frac{A_i}{s-p_i} + \sum_{k=1}^m \frac{B_k}{(s-p)^k}$$

Step 4: Expansion coefficients $A_1$ to $A_n$

$$A_i = (s-p_i)X(s)\Big|_{s=p_i}, \quad i = 1,2,\ldots,n$$

$$B_k = \left[\frac{1}{(m-k)!}\frac{d^{m-k}}{ds^{m-k}}[(s-p)^mX(s)]\right]\Bigg|_{s=p}, \quad k = 1,2,\ldots m$$

$$B_m = (s-p)^mX(s)\Big|_{s=p}$$

$$B_{m-1} = \left[\frac{d}{ds}((s-p)^mX(s))\right]\Big|_{s=p}$$

$$B_{m-2} = \left[\frac{d^2}{2ds^2}((s-p)^mX(s))\right]\Big|_{s=p}$$

Step 5: Apply the Laplace pairs

$$\frac{(n-1)!}{(s+a)^n} \longleftrightarrow t^{n-1}e^{-at}u(t)$$

with $a = -p$

$$x(t) = \mathcal{L}^{-1}[X(s)] = \left[\sum_{i=1}^n A_ie^{p_it} + \sum_{k=1}^m \frac{B_kt^{k-1}}{(k-1)!}e^{pt}\right]u(t)$$

---

## Shift property

$$x(t)u(t) \longleftrightarrow X(s)$$

$$\mathcal{L}[x(t-T)u(t-T)] = \int_{0^-}^\infty x(t-T)u(t-T)e^{-st}\,dt$$

Let $\mu = t-T$

$$\int_{0^-}^\infty x(\mu)u(\mu)e^{-s(\mu+T)}\,d\mu$$

$$= e^{-sT}\int_0^\infty x(\mu)e^{-s\mu}\,d\mu = e^{-sT}X(s)$$

delayer

This only works for $T>0$ since the signal may no longer be causal for a negative T.

---

## Frequency Shift

$$e^{-at}x(t) \longleftrightarrow X(s+a)$$

$$\mathcal{L}[e^{-at}x(t)] = \int_{0^-}^\infty e^{-at}x(t)e^{-st}\,dt$$

$$= \int_{0^-}^\infty e^{-(s+a)t}x(t)\,dt$$

$$= X(s+a)$$

we can change all instances of s with (s+a).

---

### Deriving Time Shift Property.

We might derive it on the test

Step 1: Delay by T

$$\mathcal{L}[x(t-T)u(t-T)] = \int_0^\infty x(t-T)u(t-T)e^{-st}\,dt$$

Step 2: $\mu = t-T$

$$= \int_{-T}^\infty x(\mu)u(\mu)e^{-s(\mu+T)}\,d\mu$$

delayer $\to$

$$e^{-sT}\int_{-T}^\infty x(\mu)u(\mu)e^{-s\mu}\,d\mu$$

Step 3:

$$e^{-sT}\underbrace{\int_{0^-}^\infty x(\mu)e^{-s\mu}\,d\mu}_{X(s)} = e^{-sT}X(s)$$

---

### Ex: More Complex

$$X(s) = \underbrace{\frac{4}{s+2}} + \underbrace{\frac{6}{(s+5)^2}} + \underbrace{\frac{8}{s^2+4s+5}}$$

$\downarrow 4e^{-2t} \qquad \downarrow 6te^{-5t}$

break them down

$$\frac{8}{s^2+4s+5} = \frac{8}{s^2+4s+4+1} \longrightarrow \frac{8}{(s+2)^2+1}$$

$$e^{-at}\sin(\omega_0 t)u(t) \longleftrightarrow \frac{\omega_0}{(s+a)^2+\omega_0^2}$$

$$\mathcal{L}^{-1}\left[\frac{8}{(s+2)^2+1}\right] = 8e^{-2t}\sin(t)u(t)$$

$$x(t) = [4e^{-2t} + 6te^{-5t} + 8e^{-2t}\sin(t)]u(t)$$

---

Example: $\mathcal{L}[t^2e^{-3t}\underline{\cos(4t)u(t)}]$

$$\cos(4t)u(t) = \frac{s}{s^2+4^2}$$

$$e^{-at}x(t) \longleftrightarrow X(s+a)$$

$$X(s) = e^{-3t}\cos(4t)u(t) = \frac{s+3}{(s+3)^2+4^2}$$

$$\mathcal{L}[t^2x(t)] = \frac{d^2}{ds^2}[X(s)]$$

$$\mathcal{L}[t^2e^{-3t}\cos(4t)u(t)] = \frac{d^2}{ds^2}\left[\frac{s+3}{(s+3)^2+4^2}\right]$$

---

## Poles

- when we have $\frac{1}{s}$ a pole at $s=0$, which is $u(t)$ for an impulse input.
- The output $\frac{1}{s^2}$ is a ramp function.

The system is only stable when the poles are in the Open Left Half Plane

Divison Partial Relationship $m=n$

$$\frac{N(s)}{D(s)} = 1 + \frac{N(s)-D(s)}{D(s)}$$

---

## Spectrum Notes

Sampling Rate

$$f_s > 2f_0$$

$$\omega_0 = \frac{2\pi}{T_0} \to f = 2\pi\omega_0$$

points of freq:

$$\pm f_0 + nf_s$$

Nyquist Band $= \frac{f_s}{2}$

F.S of a Pulse Train

$$\delta_{T_s}(t) = f_s\sum e^{j2\pi nf_st}$$

$$e^{j2n\pi ft} \longrightarrow \delta(f-nf_s)$$

freq. domain expression

$$\Lambda(f) = \mathcal{F}[\delta_{T_s}(t)]$$

$$= f_s\sum\delta(f-nf_s)$$

$$x(t) = x_s(t) * h(t)$$

$$x_s(t) = \sum x(nT_s)\,\delta(t-nT_s)$$

$$x(t) = \sum x(nT_s)\,\delta(t-nT_s) * h(t)$$

$$= \sum x(nT_s)\,h(t-nT_s)$$

$$= \sum x(nT_s)\frac{\sin(\pi f_s(t-nT_s))}{\pi f_s(t-nT_s)}$$

---

History:

Marquis ... de Laplace

The "Isaac Newton" of France

Sharon Story: Analog Signal to Digital

Sampling Rate:

Harry Nyquist

Claude Elwood Shannon

Parseval's theorem

1799 - Marc-Antonine Parseval

- the Power of sum of sin =

Basel Problem:

- proposed = Pietro Mengoli = 1650
- proved = Leonhard Euler = 1735

Hometown = Basel, Switzerland.

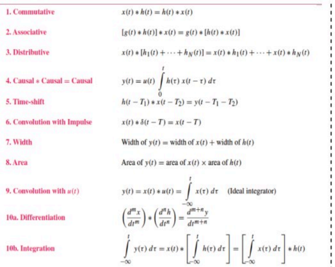
Fourier:

Jean-Baptiste Joseph Fourier

"green house effect."

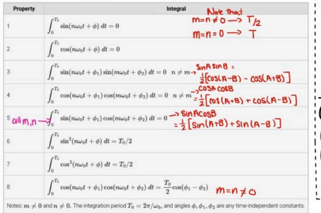
Gibbs Theorem:

Josiah Willard Gibbs - 1899



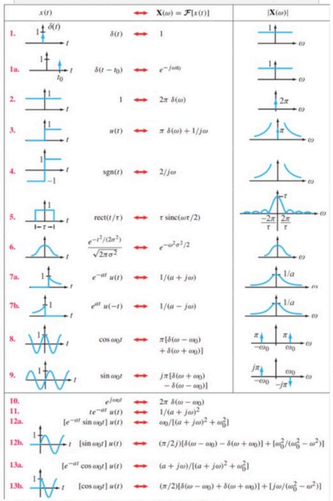
| | |
|---|---|
| **1. Commutative** | $x(t)*h(t)=h(t)*x(t)$ |
| **2. Associative** | $[g(t)*h(t)]*x(t)=g(t)*[h(t)*x(t)]$ |
| **3. Distributive** | $x(t)*[h_1(t)+\cdots+h_N(t)]=x(t)*h_1(t)+\cdots+x(t)*h_N(t)$ |
| **4. Causal * Causal = Causal** | $y(t)=u(t)\int_0^t h(\tau)\,x(t-\tau)\,d\tau$ |
| **5. Time-shift** | $h(t-T_1)*x(t-T_2)=y(t-T_1-T_2)$ |
| **6. Convolution with Impulse** | $x(t)*\delta(t-T)=x(t-T)$ |
| **7. Width** | Width of $y(t)$ = width of $x(t)$ + width of $h(t)$ |
| **8. Area** | Area of $y(t)$ = area of $x(t)$ × area of $h(t)$ |
| **9. Convolution with $u(t)$** | $y(t)=x(t)*u(t)=\int_{-\infty}^{t}x(\tau)\,d\tau$ (Ideal integrator) |
| **10a. Differentiation** | $\left(\frac{d^m x}{dt^m}\right)*\left(\frac{d^n h}{dt^n}\right)=\frac{d^{m+n}y}{dt^{m+n}}$ |
| **10b. Integration** | $\int_{-\infty}^{t}y(\tau)\,d\tau=x(t)*\left[\int_{-\infty}^{t}h(\tau)\,d\tau\right]=\left[\int_{-\infty}^{t}x(\tau)\,d\tau\right]*h(t)$ |

Table 5.4.1: Trigonometric integral properties for any integers $m$ and $n$.



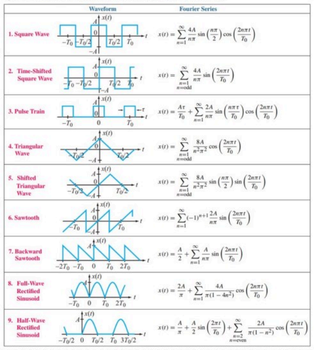
| Property | Integral |
|---|---|
| 1 | $\int_0^{T_0}\sin(n\omega_0 t+\phi)\,dt=0$ Note that $m=n\neq 0\to T/2$; $m=n=0\to T$ |
| 2 | $\int_0^{T_0}\cos(n\omega_0 t+\phi)\,dt=0$ |
| 3 | $\int_0^{T_0}\sin(n\omega_0 t+\phi_1)\sin(m\omega_0 t+\phi_2)\,dt=0$ $n\neq m \to$ $\sin A\sin B=\frac{1}{2}[\cos(A-B)-\cos(A+B)]$ |
| 4 | $\int_0^{T_0}\cos(n\omega_0 t+\phi_1)\cos(m\omega_0 t+\phi_2)\,dt=0$ $n\neq m \to$ $\cos A\cos B=\frac{1}{2}[\cos(A+B)+\cos(A-B)]$ |
| 5 | all m,n $\to$ $\int_0^{T_0}\sin(n\omega_0 t+\phi_1)\cos(m\omega_0 t+\phi_2)\,dt=0 \to$ $\sin A\cos B=\frac{1}{2}[\sin(A+B)+\sin(A-B)]$ |
| 6 | $\int_0^{T_0}\sin^2(n\omega_0 t+\phi)\,dt=T_0/2$ |
| 7 | $\int_0^{T_0}\cos^2(n\omega_0 t+\phi)\,dt=T_0/2$ |
| 8 | $\int_0^{T_0}\cos(n\omega_0 t+\phi_1)\cos(n\omega_0 t+\phi_2)\,dt=\frac{T_0}{2}\cos(\phi_1-\phi_2)$ $m=n\neq 0$ |

Notes: $m\neq 0$ and $n\neq 0$. The integration period $T_0=2\pi/\omega_0$, and angles $\phi,\phi_1,\phi_2$ are any time-independent constants.



| | $x(t)$ | ↔ $\mathbf{X}(\omega)=\mathcal{F}[x(t)]$ |
|---|---|---|
| 1. | $\delta(t)$ | $1$ |
| 1a. | $\delta(t-t_0)$ | $e^{-j\omega t_0}$ |
| 2. | $1$ | $2\pi\,\delta(\omega)$ |
| 3. | $u(t)$ | $\pi\,\delta(\omega)+1/j\omega$ |
| 4. | $\mathrm{sgn}(t)$ | $2/j\omega$ |
| 5. | $\mathrm{rect}(t/\tau)$ | $\tau\,\mathrm{sinc}(\omega\tau/2)$ |
| 6. | $\frac{e^{-t^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}$ | $e^{-\omega^2\sigma^2/2}$ |
| 7a. | $e^{-at}\,u(t)$ | $1/(a+j\omega)$ |
| 7b. | $e^{at}\,u(-t)$ | $1/(a-j\omega)$ |
| 8. | $\cos\omega_0 t$ | $\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$ |
| 9. | $\sin\omega_0 t$ | $j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$ |
| 10. | $e^{j\omega_0 t}$ | $2\pi\,\delta(\omega-\omega_0)$ |
| 11. | $te^{-at}\,u(t)$ | $1/(a+j\omega)^2$ |
| 12a. | $[e^{-at}\sin\omega_0 t]\,u(t)$ | $\omega_0/[(a+j\omega)^2+\omega_0^2]$ |
| 12b. | $[\sin\omega_0 t]\,u(t)$ | $(\pi/2j)[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]+[\omega_0^2/(\omega_0^2-\omega^2)]$ |
| 13a. | $[e^{-at}\cos\omega_0 t]\,u(t)$ | $(a+j\omega)/[(a+j\omega)^2+\omega_0^2]$ |
| 13b. | $[\cos\omega_0 t]\,u(t)$ | $(\pi/2)[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]+[j\omega/(\omega_0^2-\omega^2)]$ |

Figure 5.4.2: Fourier series expressions for a select set of periodic waveforms.



| Waveform | Fourier Series |
|---|---|
| 1. Square Wave | $x(t)=\sum_{n=1}^{\infty}\frac{4A}{n\pi}\sin\left(\frac{n\pi}{2}\right)\cos\left(\frac{2n\pi t}{T_0}\right)$ |
| 2. Time-Shifted Square Wave | $x(t)=\sum_{n=1,\ n=\text{odd}}^{\infty}\frac{4A}{n\pi}\sin\left(\frac{2n\pi t}{T_0}\right)$ |
| 3. Pulse Train | $x(t)=\frac{A\tau}{T_0}+\sum_{n=1}^{\infty}\frac{2A}{n\pi}\sin\left(\frac{n\pi\tau}{T_0}\right)\cos\left(\frac{2n\pi t}{T_0}\right)$ |
| 4. Triangular Wave | $x(t)=\sum_{n=1,\ n=\text{odd}}^{\infty}\frac{8A}{n^2\pi^2}\cos\left(\frac{2n\pi t}{T_0}\right)$ |
| 5. Shifted Triangular Wave | $x(t)=\sum_{n=1,\ n=\text{odd}}^{\infty}\frac{8A}{n^2\pi^2}\sin\left(\frac{n\pi}{2}\right)\sin\left(\frac{2n\pi t}{T_0}\right)$ |
| 6. Sawtooth | $x(t)=\sum_{n=1}^{\infty}(-1)^{n+1}\frac{2A}{n\pi}\sin\left(\frac{2n\pi t}{T_0}\right)$ |
| 7. Backward Sawtooth | $x(t)=\frac{A}{2}+\sum_{n=1}^{\infty}\frac{A}{n\pi}\sin\left(\frac{2n\pi t}{T_0}\right)$ |
| 8. Full-Wave Rectified Sinusoid | $x(t)=\frac{2A}{\pi}+\sum_{n=1}^{\infty}\frac{4A}{\pi(1-4n^2)}\cos\left(\frac{2n\pi t}{T_0}\right)$ |
| 9. Half-Wave Rectified Sinusoid | $x(t)=\frac{A}{\pi}+\frac{A}{2}\sin\left(\frac{2\pi t}{T_0}\right)+\sum_{n=2,\ n=\text{even}}^{\infty}\frac{2A}{\pi(1-n^2)}\cos\left(\frac{2n\pi t}{T_0}\right)$ |

## Time Derivative

$x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{X}(\omega)e^{j\omega t}\,d\omega$

differentiating both sides

$\frac{dx(t)}{dt}=\frac{1}{2\pi}\int_{-\infty}^{\infty}j\omega\hat{X}(\omega)e^{j\omega t}\,d\omega$

$\frac{dx(t)}{dt}\longleftrightarrow j\omega\hat{X}(\omega)$

## Time Shift Property

$\mathcal{F}[x(t-T)]=\int_{-\infty}^{\infty}x(t-T)e^{-j\omega t}\,dt$

Let $u=t-T$  $du=dt$

$=\int_{-\infty}^{\infty}x(u)e^{-j\omega(u+T)}\,du=e^{-j\omega T}\int_{-\infty}^{\infty}x(u)e^{-j\omega u}\,du=e^{-j\omega T}\hat{X}(\omega)$

## Freq. Shift Property

$\mathcal{F}[e^{j\omega_0 t}x(t)]=\int_{-\infty}^{\infty}x(t)e^{j\omega_0 t}\cdot e^{-j\omega t}\,dt$

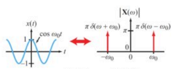
$=\int_{-\infty}^{\infty}x(t)e^{-j(\omega-\omega_0)t}\,dt=\hat{X}(\omega-\omega_0)$

$e^{j\omega_0 t}x(t)\longleftrightarrow\hat{X}(\omega-\omega_0)$

## Fourier Transform of Cosine + Sine

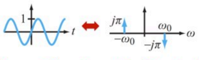
$\cos(\omega_0 t)=\frac{e^{j\omega_0 t}+e^{-j\omega_0 t}}{2}$

$\to X(\omega)=\pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0)$



$\sin(\omega_0 t)=\frac{e^{j\omega_0 t}-e^{-j\omega_0 t}}{2j}=\frac{-j}{2}\left(e^{j\omega_0 t}-e^{-j\omega_0 t}\right)$

$=X(\omega)=j\pi\delta(\omega-\omega_0)+j\pi\delta(\omega+\omega_0)$



## Fourier Transform of a Periodic Signal

$x(t)=\sum_{n=-\infty}^{\infty}X_n e^{jn\omega_0 t}$

we can get this:

$\mathcal{F}[x(t)]=\sum_{n=-\infty}^{\infty}X_n\mathcal{F}[e^{jn\omega_0 t}]=\sum_{n=-\infty}^{\infty}X_n 2\pi\delta(\omega-n\omega_0)$

the fourier transform of a period signal is a series of impulses, one impulse for each harmonic (a pulse train)

---

## Fourier Coefficients

$x(t)\cos(\omega t)\longleftarrow$ even (always)

**Even Symmetry:** $x(t)=x(-t)$

$a_0=\frac{2}{T_0}\int_0^{\frac{T_0}{2}}x(t)\,dt$

$a_n=\frac{4}{T_0}\int_0^{\frac{T_0}{2}}x(t)\cos(n\omega_0 t)\,dt$

$b_n=0$  $C_n=|a_n|$

$\phi_n=\begin{cases}0 & \text{if } a_n>0\\ 180^\circ & \text{if } a_n<0\end{cases}$

$x(t)\sin(\omega t)\longleftarrow$ odd (always)

**Odd Symmetry:** $x(t)=-x(-t)$

$a_0=0$, $a_n=0$

$b_n=\frac{4}{T_0}\int_0^{\frac{T_0}{2}}x(t)\sin(n\omega_0 t)\,dt$

$C_n=|b_n|$, $\phi_n=\begin{cases}-90 & \text{if } b_n>0\\ 90 & \text{if } b_n<0\end{cases}$

$x(t)=a_0+\sum[a_n\cos(n\omega_0 t)+b_n\sin(n\omega_0 t)]$

$\cos(n\omega_0 t)=\frac{1}{2}\left[e^{jn\omega_0 t}+e^{-jn\omega_0 t}\right]$

$\sin(n\omega_0 t)=\frac{1}{2j}\left[e^{jn\omega_0 t}-e^{-jn\omega_0 t}\right]$ > Euler's Formula

$x(t)=a_0+\sum\left[\underbrace{\frac{a_n}{2}\left(e^{jn\omega_0 t}+e^{-jn\omega_0 t}\right)}_{\text{Even}}+\underbrace{\frac{b_n}{2}\left(e^{jn\omega_0 t}-e^{-jn\omega_0 t}\right)}_{\text{Odd}}\right]$

$=a_0+\sum\left[\frac{(a_n-jb_n)}{2}e^{jn\omega_0 t}+\frac{(a_n+jb_n)}{2}e^{-jn\omega_0 t}\right]$

This turns into

$=a_0+\sum[X_n e^{jn\omega_0 t}+X_{-n}e^{-jn\omega_0 t}]$  $X_{-n}=\frac{(a_n+jb_n)}{2}=X_n^*$ conjugate

$=\sum_{n=-\infty}^{\infty}X_n e^{jn\omega_0 t}$  $X_0=a_0$

and from this we can get:  In this we can see that n=m, the exponential is 1

$\int_{-T_0/2}^{T_0/2}x(t)e^{-jm\omega_0 t}\,dt=\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}}\sum X_n e^{jn\omega_0 t}e^{-jm\omega_0 t}\,dt$

$=\sum_{n=-\infty}^{\infty}\int_{-T_0/2}^{T_0/2}X e^{j(n-m)\omega_0 t}\,dt=T_0 X_n$

understand that $X_n$ in this case is.

$\frac{1}{T_0}\int_{-T_0/2}^{T_0/2}x(t)e^{-jn\omega_0 t}\,dt$

## Fourier Series

Case 1: $a_n>0$, $b_n<0\longrightarrow\phi_n=-\tan^{-1}\left(\frac{b_n}{a_n}\right)$

Cas 2: $a_n<$ , $b_n<0\longrightarrow\phi_n=\pi-\tan^{-1}\left(\frac{b_n}{a_n}\right)$

Case 3: $a_n<$ , $b_n>0\longrightarrow\phi_n=\pi-\tan^{-1}\left(\frac{b_n}{a_n}\right)$

Case4: $a_n$ 0, $b_n>0\longrightarrow\phi_n$ $-\tan^{-1}\left(\frac{b_n}{a_n}\right)$

sin(nω0t) = odd  cos(nω0t): even

$a_n=\frac{2}{T_0}\int_{T_0}x(t)\cos(n\omega_0 t)\,dt$

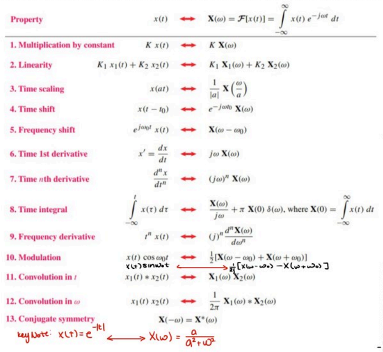
odd × even = odd → $a_n=0$

$b_n=\frac{2}{T_0}\int_{T_0}x(t)\sin(n\omega_0 t)\,dt$

even × odd = odd → $b_n=0$

$\mathrm{sinc}=\begin{cases}1 & \text{for } x=0\\ \frac{\sin(x)}{x} & \text{otherwise}\end{cases}$

$\lim_{x\to 0}\left\{\frac{\sin(x)}{x}\right\}=1$



| Property | $x(t)$ | ↔ $\mathbf{X}(\omega)=\mathcal{F}[x(t)]=\int_{-\infty}^{\infty}x(t)\,e^{-j\omega t}\,dt$ |
|---|---|---|
| 1. Multiplication by constant | $K\,x(t)$ | $K\,\mathbf{X}(\omega)$ |
| 2. Linearity | $K_1\,x_1(t)+K_2\,x_2(t)$ | $K_1\,\mathbf{X}_1(\omega)+K_2\,\mathbf{X}_2(\omega)$ |
| 3. Time scaling | $x(at)$ | $\frac{1}{\lvert a\rvert}\mathbf{X}\left(\frac{\omega}{a}\right)$ |
| 4. Time shift | $x(t-t_0)$ | $e^{-j\omega t_0}\,\mathbf{X}(\omega)$ |
| 5. Frequency shift | $e^{j\omega_0 t}\,x(t)$ | $\mathbf{X}(\omega-\omega_0)$ |
| 6. Time 1st derivative | $x'=\frac{dx}{dt}$ | $j\omega\,\mathbf{X}(\omega)$ |
| 7. Time $n$th derivative | $\frac{d^n x}{dt^n}$ | $(j\omega)^n\,\mathbf{X}(\omega)$ |
| 8. Time integral | $\int_{-\infty}^{t}x(\tau)\,d\tau$ | $\frac{\mathbf{X}(\omega)}{j\omega}+\pi\,\mathbf{X}(0)\,\delta(\omega)$, where $\mathbf{X}(0)=\int_{-\infty}^{\infty}x(t)\,dt$ |
| 9. Frequency derivative | $t^n\,x(t)$ | $(j)^n\frac{d^n\mathbf{X}(\omega)}{d\omega^n}$ |
| 10. Modulation | $x(t)\cos\omega_0 t$<br>$x(t)\sin\omega_0 t$ | $\frac{1}{2}[\mathbf{X}(\omega-\omega_0)+\mathbf{X}(\omega+\omega_0)]$<br>$\frac{1}{2j}[X(\omega-\omega_0)-X(\omega+\omega_0)]$ |
| 11. Convolution in $t$ | $x_1(t)*x_2(t)$ | $\mathbf{X}_1(\omega)\,\mathbf{X}_2(\omega)$ |
| 12. Convolution in $\omega$ | $x_1(t)\,x_2(t)$ | $\frac{1}{2\pi}\mathbf{X}_1(\omega)*\mathbf{X}_2(\omega)$ |
| 13. Conjugate symmetry | | $\mathbf{X}(-\omega)=\mathbf{X}^*(\omega)$ |

key Note: $x(t)=e^{-|t|}\longleftrightarrow X(\omega)=\frac{a}{a^2+\omega^2}$

## The Unit Step

$x(t)=u(t)$

$\mathcal{F}[x(t)]=\int_0^{\infty}u(t)e^{-j\omega t}\,dt$

$=\left(\frac{1}{-j\omega}\right)e^{-j\omega t}\Big|_0^{\infty}=\frac{j}{\omega}\left[e^{-j\infty}-1\right]$ This doesn't converge

Approach 1:

$\mathcal{F}[e^{-\varepsilon t}u(t)]=\frac{1}{\varepsilon+j\omega}$

$=\mathcal{F}[u(t)]=\lim_{\varepsilon\to 0}\left[\frac{1}{\varepsilon+j\omega}\right]=\lim_{\varepsilon\to 0}\left[\frac{(\varepsilon-j\omega)}{(\varepsilon+j\omega)(\varepsilon-j\omega)}\right]$

$=\lim_{\varepsilon\to 0}\left[\frac{\varepsilon-j\omega}{\varepsilon^2+\omega^2}\right]=\lim_{\varepsilon\to 0}\left[\frac{\varepsilon}{\varepsilon^2+\omega^2}-\frac{j\omega}{\varepsilon^2+\omega^2}\right]$

$\frac{\varepsilon}{\varepsilon^2+\omega^2}=\frac{1}{\varepsilon}$, but since $\varepsilon\to 0$ is infinitely tall

But under the curve the area is: $\int_{-\infty}^{\infty}\frac{\varepsilon}{\varepsilon^2+\omega^2}\,d\omega=\tan^{-1}\left(\frac{\omega}{\varepsilon}\right)\Big|_{-\infty}^{\infty}=\pi$

Regardless the value of ε becomes an impulse in the limit as $\varepsilon\to 0$

$\lim_{\varepsilon\to 0}\frac{\varepsilon}{\varepsilon^2+\omega^2}=\pi\delta(\omega)$  $=\pi\delta(\omega)-\frac{j}{\omega}$

$=\pi\delta(\omega)+\frac{1}{j\omega}$

Approach #2

$\mathrm{sgn}(t)=u(t)-u(-t)$

$u(t)=\frac{1}{2}(1+\mathrm{sgn}(t))$

$\mathcal{F}[\mathrm{sgn}(t)]=\int_{-\infty}^{\infty}\lim_{\varepsilon\to 0}\left[e^{-\varepsilon t}u(t)-e^{\varepsilon t}u(-t)\right]e^{-j\omega t}\,dt$

$=\lim\left[\int_0^{\infty}e^{-(\varepsilon+j\omega)t}\,dt-\int_{-\infty}^{0}e^{(\varepsilon-j\omega)t}\,dt\right]$

$=\lim_{\varepsilon\to 0}\left[\frac{e^{-(\varepsilon+j\omega)t}}{-(\varepsilon+j\omega)}\Big|_0^{\infty}-\frac{e^{(\varepsilon-j\omega)t}}{(\varepsilon-j\omega)}\Big|_{-\infty}^{0}\right]$

$=\lim_{\varepsilon\to 0}\left[\frac{1}{\varepsilon+j\omega}-\frac{1}{\varepsilon-j\omega}\right]=\frac{2}{j\omega}$

$\mathcal{F}[u(t)]=\mathcal{F}\left[\frac{1}{2}(1+\mathrm{sgn}(t))\right]=\pi\delta(\omega)+\frac{1}{j\omega}$

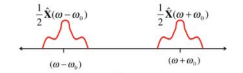
Remember: $1\longleftrightarrow 2\pi\delta(\omega)$

$\frac{d}{dt}\left[\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)e^{j\omega t}\,d\omega\right]=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)\frac{d}{dt}\left[e^{j\omega t}\right]d\omega$

$=\frac{1}{2\pi}\int_{-\infty}^{\infty}F(\omega)j\omega e^{j\omega t}\,d\omega\to\frac{1}{2\pi}(j\omega)F(\omega)e^{j\omega t}\,d\omega$

## The Spectrum of the Modulated Signal

$x(t)\cos(\omega_0 t)\longleftrightarrow\frac{1}{2}[\hat{X}(\omega-\omega_0)+\hat{X}(\omega+\omega_0)]$ modulation property



$x(t)=\frac{1}{2}\int_{-\infty}^{\infty}X(\omega)e^{j\omega t}\,d\omega$

$=x(t)\cos(\omega_0 t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\cos(\omega_0 t)X(\omega_d)e^{j\omega_d t}\,d\omega_d$

$=x(t)\cos(\omega_0 t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{e^{j\omega_0 t}+e^{-j\omega_0 t}}{2}\right)X(\omega_d)e^{j\omega_d t}\,d\omega_d$

$x(t)\cos(\omega_0 t)=\frac{1}{4\pi}\left[\int_{-\infty}^{\infty}X(\omega_d)e^{j(\omega_d+\omega_0)t}\,d\omega_d+\int_{-\infty}^{\infty}X(\omega_d)e^{j(\omega_d-\omega_0)t}\,d\omega_d\right]$

$x(t)\cos(\omega_0 t)\longleftrightarrow\frac{1}{2}[X(\omega-\omega_0)+X(\omega+\omega_0)]$

---

## Fourier Coefficients

Cosine / Sine

$a_0=\frac{1}{T_0}\int_0^{T_0}x(t)\,dt$

$a_n=\frac{2}{T_0}\int_0^{T_0}x(t)\cos(n\omega_0 t)\,dt$

$b_n=\frac{2}{T_0}\int_0^{T_0}x(t)\sin(n\omega_0 t)\,dt$

Amplitute / Phase

$x(t)=C_0+\sum_{n=1}C_n\cos(n\omega_0 t+\phi_n)$

$C_n e^{j\phi_n}=a_n-jb_n$

$C_n=\sqrt{a_n^2-jb_n}$

$\phi=\begin{cases}-\tan^{-1}\left(\frac{b_n}{a_n}\right), & a_n>0\\ \pi-\tan^{-1}\left(\frac{b_n}{a_n}\right), & a_n<0\end{cases}$

Exponential

$x(t)=\sum_{n=-\infty}X_n e^{jn\omega_0 t}$  $a_0=C_0=X_0$  $a_n=C_n\cos\phi_n$

$X_n=|X_n|e^{j\phi_n}$; $X_{-n}=X_n^*$; $\phi_{-n}=-\phi_n$  $b_n=-C_n\sin\phi_n$

$|X_n|=\frac{C_n}{2}$; $X_0=C_0$  $X_n=\frac{1}{2}(a_n-jb_n)$

$X_n=\frac{1}{T_0}\int_0^{T_0}x(t)e^{-jn\omega_0 t}\,dt$

## Pav's Theorem

$x(t)=A\cos(n\omega_0 t)\longrightarrow\omega_0\neq 0$, $n>0$

$P_{x_1}=\frac{1}{T_0}\int_0^{T_0}|A\cos(n\omega_0 t)|^2\,dt$

$=\frac{A^2}{T_0}\int_0^{T_0}\cos^2(n\omega_0 t)\,dt$

$=\frac{A^2}{T_0}\int_0^{T_0}\left[\frac{1}{2}(1+\cos 2(n\omega_0 t))\right]dt=\frac{A^2}{2}$

$P_{av}=\frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}}|x(t)|^2\,dt=\frac{1}{T_0}\int_0^{T_0}|x(t)|^2\,dt$

**Trig. Identities**

$e^{j(A+B)}=\cos(A+B)+j\sin(A+B)$

$e^{jA}e^{jB}=(\cos A+j\sin A)\times(\cos B+j\sin B)$

$=(\cos A\cos B-\sin A\sin B)+j(\sin A\cos B+\cos A\sin B)$

This can correspond:

$\cos(A+B)=\cos A\cos B-\sin A\sin B$

$\sin(A+B)=\sin A\cos B+\cos A\sin B$

$\cos A\cos B=\frac{1}{2}[\cos(A+B)+\cos(A-B)]$

$\sin A\sin B=\frac{1}{2}[\cos(A-B)-\cos(A+B)]$

$\sin A\cos B=\frac{1}{2}[\sin(A+B)+\sin(A-B)]$

other ident.

$\sin(x\pm y)=\sin x\cos y\pm\cos x\sin y$

$\cos(x\pm y)=\cos x\cos y\mp\sin x\sin y$

$\sin(2x)=2\sin x\cos x$

$\cos(2x)=\cos^2 x-\sin^2 x$

$\sin x\sin y=\frac{1}{2}[\cos(x-y)-\cos(x+y)]$

$\cos x\cos y=\frac{1}{2}[\cos(x-y)+\cos(x+y)]$

$\sin x=\pm\cos(x\mp 90)$

$\cos x=\pm\sin(x\pm 90)$

$\sin x=-\sin(x\pm 180)$

$\cos x=-\cos(x\pm 180)$

$\sin(-x)=-\sin x$

$\cos(-x)=\cos x$

$\sin^2 x=\frac{1}{2}(1-\cos 2x)$

$\cos^2 x=\frac{1}{2}(1+\cos 2x)$

## Euler's Relationship Example

$\sin(2x)=2\sin x\cos x$

Since $2x=x+x$ we need to sqr the eqn.

$\left(e^{j2x}\right)^2=\left(\cos 2x+\sin 2x\right)^2$

$e^{jx}=\cos x+\sin x\longrightarrow e^{j2x}=(\cos x+j\sin x)^2$ expand this

$(\cos x+j\sin x)(\cos x+j\sin x)$

$\cos x^2+2j\cos x\sin x+j^2\sin^2 x$

$=\cos^2 x-\sin^2 x+j(2\sin x\cos x)$

$\cos 2x+j\sin 2x=(\cos^2 x-\sin^2 x)+j(2\sin x\cos x)$

this will follow the $A+jB=C+jD$ rule, where $A=C$ + $jB=jD$

$\cos 2x=\cos^2 x-\sin^2 x$

$\boxed{\sin 2x=2\sin x\cos x}$

## Decaying Exponentials

$x(t)=e^{-at}u(t)$  $a>0$

$\mathcal{F}[x(t)]=\int_{-\infty}^{\infty}e^{-at}u(t)e^{-j\omega t}\,dt$

$=\int_0^{\infty}e^{-at}e^{-j\omega t}\,dt=\left(\frac{-1}{(a+j\omega)}e^{-(a+j\omega)t}\right)\Big|_{t=0}^{t\to\infty}$

$=\frac{1}{a+j\omega}$

$y(t)=\int_{-\infty}^{\infty}x(z)h(t-z)\,dz\to\int_{-\infty}^{\infty}y(t)e^{-j\omega t}\,dt\to\int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty}x(z)h(t-z)\,dz\right)e^{-j\omega t}\,dt$

$=\int_{-\infty}^{\infty}x(z)\left(\int_{-\infty}^{\infty}h(t-z)e^{-j\omega t}\right)dz$  usub: $\int_{-\infty}^{\infty}x(z)\left(\int_{-\infty}^{\infty}x(u)e^{-j\omega t}\,du\right)dz$

$u=t-z$  $du=dt$

$t=u+z$

$=\underbrace{\int_{-\infty}^{\infty}x(z)e^{-j\omega t}\,dz}_{X(\omega)}\underbrace{\int_{-\infty}^{\infty}h(u)e^{-j\omega u}\,du}_{H(\omega)}$

## Fourier Transform of the Modulated Signal

$\mathcal{F}[x(t)\cos(\omega_0 t)]=\mathcal{F}\left[x(t)\frac{(e^{j\omega_0 t}+e^{-j\omega_0 t})}{2}\right]$

$=\frac{1}{2}[\underbrace{\hat{X}(\omega-\omega_0)}_{\text{Low Freq.}}+\underbrace{\hat{X}(\omega+\omega_0)}_{\text{High Freq.}}]$

a) $X(\omega)=\int_{-\infty}^{\infty}x(t)e^{-j\omega t}\,dt$

$e^{-j\omega t}\longleftrightarrow 2\pi\delta(\omega-\omega_0)$

$x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}X(\omega)e^{j\omega t}\,dt$

for $X(\omega)$ to be $2\pi\delta(\omega-\omega_0)$

$x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}2\pi\delta(\omega-\omega_0)e^{j\omega t}\,d\omega=e^{j\omega_0 t}$

$\boxed{\mathcal{F}[e^{j\omega_0 t}]=2\pi\delta(\omega-\omega_0)}$

$x(t)=\overset{\text{Odd}}{\sin(t)}\to-\sin(t)$

$x(t)=\overset{\text{even}}{\cos(t)}\longrightarrow\cos(-t)$

$x(t)=\overset{\text{odd}}{\tan(t)}\longrightarrow-\tan(t)$

$x(t)=|t|\xrightarrow{\text{even}}|t|$

Delayer:

$x(t)*\delta(t-T)=x(t-T)$

Integrator:

$x(t)*u(t)=\int_{-\infty}^{t}x(t)\,dt$

unit step:

$u(t)*u(t)=r(t)$

Ramp:

$r(t)*u(t)=\frac{1}{2}t^2u(t)$

B) $te^{-at}u(t)\longleftrightarrow\frac{1}{(a+j\omega)^2}$ (for $a>0$)

First we need the Fourier of $e^{-at}u(t)$ because its a step function

$\mathcal{F}[e^{-at}u(t)]=\int_0^{\infty}e^{-at}e^{-j\omega t}\,dt=\int_0^{\infty}e^{-(a+j\omega)t}\,dt$

$\left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)}\right]_0^{\infty}\longrightarrow\frac{1}{a+j\omega}$

Now we need to find the Fourier of this equation with respect

$\mathcal{F}[tf(t)]=j\frac{dF(\omega)}{d\omega}$

$t\underbrace{e^{-at}u(t)}_{f(t)}\to$

$j\cdot\frac{d}{d\omega}\left(\frac{1}{a+j\omega}\right)=\frac{-j}{(a+j\omega)^2}$

this is seperate from the $\frac{d}{d\omega}$

$j\left(\frac{-j}{(a+j\omega)^2}\right)=\frac{1}{(a+j\omega)^2}$

$\boxed{\mathcal{F}[te^{-at}u(t)]=\frac{1}{(a+j\omega)^2}}$

C) Euler's formula: $\sin(\omega_0 t)=\frac{e^{j\omega_0 t}-e^{-j\omega_0 t}}{2j}$

or Modulation: $\frac{1}{2j}(X(\omega-\omega_0)-Y(\omega+\omega_0))$

$[e^{-at}\sin(\omega_0 t)]u(t)\longleftrightarrow\frac{\omega_0}{(a+j\omega)^2+\omega_0^2}$

$e^{-at}\sin(\omega_0 t)u(t)\to\frac{1}{2j}\left[e^{-at}e^{j\omega_0 t}-e^{-at}e^{-j\omega_0 t}\right]u(t)$

This is also modulation $\frac{1}{2j}\left(e^{-(a-j\omega_0)t}-e^{-(a+j\omega_0)t}\right)u(t)$

$\mathcal{F}[e^{-at}u(t)]=\frac{1}{a+j\omega}$ ← plug this into the equation

$\frac{1}{2j}\left[\frac{1}{a+j(\omega-\omega_0)}-\frac{1}{a+j(\omega+\omega_0)}\right]$

$=\frac{1}{2j}\left[\frac{a+j(\omega+\omega_0)-a+j(\omega-\omega_0)}{[a+j(\omega-\omega_0)][a+j(\omega+\omega_0)]}\right]$

$\frac{1}{\cancel{2j}}\left[\frac{\cancel{j(2\omega_0)}}{(a+j\omega)^2+\omega_0^2}\right]\to\frac{\omega_0}{(a+j\omega)^2+\omega_0^2}$

$\boxed{\mathcal{F}[e^{-at}\sin(\omega_0 t)u(t)]=\frac{\omega_0}{(a+j\omega)^2+\omega_0^2}}$

D) $\sin(\omega_0 t)u(t)\longleftrightarrow\frac{\pi}{2}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]+\frac{\omega_0}{\omega_0^2-\omega^2}$

We are gonna use the sgn function for this!

Recall: the relationship between unit step + signum

$u(t)=\frac{1}{2}+\frac{1}{2}\mathrm{sgn}(t)$;  $\mathrm{sgn}(t)=\begin{cases}1, & t>0\\ -1, & t<0\\ 0, & t=0\end{cases}$

Step 1. $\sin(\omega_0 t)u(t)=\frac{1}{2}\sin(\omega_0 t)+\frac{1}{2}\sin(\omega_0 t)\mathrm{sgn}(t)$

Step 2. Use Euler's or Modulation to get the Fourier Transform.

Euler's Formula: $\sin(\omega_0 t)=\frac{e^{j\omega_0 t}-e^{-j\omega_0 t}}{2j}$

Modulation: $\sin(\omega_0 t)=\frac{1}{2j}(\delta(\omega-\omega_0)-\delta(\omega+\omega_0))$

We can get the Fourier of $\sin(\omega_0 t)$ is

$\mathcal{F}[\sin(\omega_0 t)]=\frac{1}{\cancel{2}j}\left[\cancel{2}\pi\delta(\omega-\omega_0)-\cancel{2}\pi\delta(\omega+\omega_0)\right]$

this gets you the first half of the eqn $\longrightarrow\frac{\pi}{j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$

Step 3. Fourier Transform of $\sin(\omega_0 t)\mathrm{sgn}(t)$

this will solve for the othe half of the eqn

$\sin(\omega_0 t)\mathrm{sgn}(t)=\frac{1}{2j}\left[e^{j\omega_0 t}\mathrm{sgn}(t)-e^{-j\omega_0 t}\mathrm{sgn}(t)\right]$

If $\mathcal{F}[\mathrm{sgn}(t)]=\frac{2}{j\omega}$

$\mathcal{F}[\sin(\omega_0 t)\mathrm{sgn}(t)]=\frac{1}{2j}\left[\frac{2}{j(\omega-\omega_0)}-\frac{2}{j(\omega+\omega_0)}\right]$

$\frac{1}{2j}\cdot\frac{2}{j}=\frac{1}{j^2}\to\frac{1}{(-1)}=-1$

$=-\left[\frac{1}{\omega-\omega_0}-\frac{1}{\omega+\omega_0}\right]\to\left[\frac{(\omega+\omega_0)-(\omega-\omega_0)}{\omega^2-\omega_0^2}\right]=\frac{-2\omega_0^2}{\omega^2-\omega_0^2}$

$\mathcal{F}[\sin(\omega_0 t)\mathrm{sgn}(t)]=\frac{2\omega_0^2}{\omega_0^2-\omega^2}$

Step 4: Combine and plug

$\frac{1}{2}\sin(\omega_0 t)+\frac{1}{2}\sin(\omega_0 t)\mathrm{sgn}(t)$

$\boxed{\mathcal{F}[\sin(\omega_0 t)u(t)]=\frac{\pi}{2j}[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]+\frac{\omega_0^2}{\omega^2-\omega_0^2}}$

## $[e^{-at}\cos(\omega_0 t)]u(t)$

$\mathcal{F}[e^{-at}u(t)]=\frac{1}{a+j\omega}$

Euler's Formula: $\cos(\omega_0 t)=\frac{e^{j\omega_0 t}-e^{-j\omega_0 t}}{2}$

$e^{-at}\cos(\omega_0 t)u(t)=\frac{1}{2}\left[\underset{\frac{1}{a-j(\omega-\omega_0)}}{e^{-at}e^{j\omega_0 t}u(t)}+\underset{\frac{1}{a+j(\omega+\omega_0)}}{e^{-at}e^{-j\omega_0 t}u(t)}\right]$

$\to\frac{1}{2}\left[\frac{1}{a+j(\omega-\omega_0)}-\frac{1}{a-j(\omega+\omega_0)}\right]$

$\frac{1}{\cancel{2}}\left[\frac{\cancel{2}a+j2\omega}{a^2+j2a\omega+(j\omega)^2-\omega_0^2}\right]\to\frac{a+j\omega}{(a+j\omega)^2+\omega_0^2}$

$\boxed{\mathcal{F}[e^{-at}\cos(\omega_0 t)u(t)]=\frac{a+j\omega}{(a+j\omega)^2+\omega_0^2}}$

## $[\cos(\omega_0 t)]u(t)$

Euler's Formula: $\cos(\omega_0 t)u(t)=\left[\frac{e^{j\omega_0 t}-e^{-j\omega_0 t}}{2}\right]u(t)$

$\mathcal{F}[u(t)]=\pi\delta(\omega)+\frac{1}{j\omega}$; $\mathcal{F}[e^{j\omega_0 t}u(t)]=\pi\delta(\omega-\omega_0)+\frac{1}{j(\omega-\omega_0)}$

$\mathcal{F}[e^{-j\omega_0 t}u(t)]=\pi\delta(\omega+\omega_0)+\frac{1}{j(\omega+\omega_0)}$

$\mathcal{F}[\cos(\omega_0 t)u(t)]=\frac{1}{2}\left[\pi\delta(\omega-\omega_0)+\frac{1}{j(\omega-\omega_0)}+\pi\delta(\omega+\omega_0)+\frac{1}{j(\omega+\omega_0)}\right]$

$=\frac{\pi}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]+\frac{1}{2j}\left[\frac{1}{\omega-\omega_0}+\frac{1}{\omega+\omega_0}\right]$

$\frac{1}{2j}\left[\frac{(\omega-\omega_0)+(\omega+\omega_0)}{\omega^2-\omega_0^2}\right]\to\frac{\omega}{j(\omega^2-\omega_0^2)}=\frac{j\omega}{\omega^2-\omega_0^2}$

$\boxed{\mathcal{F}[\cos(\omega_0 t)u(t)]=\frac{\pi}{2}[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]+\frac{j\omega}{\omega_0^2-\omega^2}}$

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**A+jB = C+jD Rule**

$A=C$ + $jB=jD$

Orthogonality:

$\frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}}e^{j(n-m)\omega_0 t}\,dt=\begin{cases}1, & n=m\\ 0, & n\neq m\end{cases}$

Cosine / Sine Version:

$\int_{-T_0/2}^{T_0/2}\cos(n\omega_0 t)\cos(n\omega_0 t)\,dt=\int_{-T_0/2}^{T_0/2}\sin(n\omega_0 t)\sin(n\omega_0 t)\,dt=\begin{cases}T_0/2, & n=m\neq 0\\ 0, & n\neq m\end{cases}$

$\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}}\sin(n\omega_0 t)\cos(n\omega_0 t)\,dt=0$ always = 0

IBP: $uv-\int v\,du$

chain Rule:

$\frac{d}{dx}\left[(f(x))^n\right]=n(f(x))^n\cdot f'(x)$

or

$\frac{d}{dx}\left[f(g(x))\right]$

$=f'(g(x))\cdot g'(x)$

Convolutions $y(t)=x(t)*h(t)=\int_{-\infty}^{\infty}x(z)h(t-z)\,dz$

Shift: $y(t)=x(t-T)$

Scaling: $y(t)=x(at)$

Reversal: $y(t)=x(-t)$

Combined: $y(t)=x(at-b)$

If $|a|<1$ = expand

If $|a|>1$ = compress

If $|a|<0$ = Reflect

↳ Combined Transformations

method 1

$x(t)\xrightarrow{\text{time scale}}x(at)\xrightarrow{\text{shift}}x\left(a\left(t-\frac{b}{a}\right)\right)$

method 2

$x(t)\xrightarrow{\text{shift}}x(t-b)\xrightarrow{\text{time scale}}x(at-b)$

Commutative: $x*h=h*x$

Associative:

$x*(h_1*h_2)=(x*h_1)*h_2$

Distributive:

$x*(h_1+h_2)=x*h_1+x*h_2$

Identity:

$x(t)*\delta(t-t_0)=x(t-t_0)$

$\mathcal{L}[\ ]$ + $\mathcal{F}[\ ]$:

$\mathcal{L}[x*h]=X(s)H(s)$

$\mathcal{F}[x*h]=X(\omega)H(\omega)$

**Deriv. + Anti Deriv**

$\frac{d}{dx}[\sin x]=\cos x$  $\int\cos dx=\sin x+C$

$\frac{d}{dx}[\cos x]=-\sin x$  $\int\sin dx=-\cos x+C$

Extra Symmetry Notes

Even: $x(-t)=x(t)$

cosine: Only $a_n$ terms

sine: No sine terms

Exp: $X_{-n}=X_n$ (Real)

Odd: $x(-t)=-x(t)$

cosine: No cosine terms

sine: Only $b_n$ terms

Exp: $X_{-n}=-X_n$ (imaginary)